

Group Chat Influence and Linguistic Innovation: Are the Word-Coiners Also the Conversation-Leaders?

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Abstract

This project examines whether users who originate and spread group-specific vocabulary within a small, long-running group chat are also the users who exert the greatest conversational influence. Using a 15-month export of a nine-member iMessage group chat (25,572 messages), candidate neologisms were identified by filtering against a standard English dictionary and manually pruning typos. Four novel linguistic metrics (creativity, influence, zeal, and fortitude) were defined to characterize each user's role in coining and spreading new terms, and four conversational leadership metrics (reply-worthiness, citations, cold-starts, and reaction-causing) were computed from message-level features. Pearson correlations between the two metric families reveal that linguistic innovation and conversational leadership are largely dissociated: the only statistically significant relationship, between citations and linguistic influence ($r = -0.689$, $p = 0.028$), is negative, and is driven substantially by a single extreme user. These results suggest that successful lexical innovation in this community does not require – and may even be inversely related to – being a frequently cited conversational leader.

Keywords: computational sociolinguistics; group chat analysis; neologism diffusion; social influence; computational social science

1. Introduction

The goal of this project is to analyze the relationship between topic leadership and linguistic innovation within a small group chat. Specifically, I am interested in understanding whether the users who create and spread new language – neologisms – are also the ones who exert a high degree of influence over conversations.

The dataset consists of a real iMessage group chat among a group of college-aged friends. The primary focus is on two aspects of conversational behavior: (1) the origin and diffusion of group-specific vocabulary, and (2) measurable indicators of conversational control such as replies, mentions, and reactions.

This project is exploratory in nature. Rather than building a predictive model, the objective is to quantify these behaviors and examine correlations between them. This is partially motivated by a curiosity about the mechanisms and drivers of lexical innovation. Are new words that enter broader lexicons first born in hyperlocal familects and sociolects such as the one spoken in the group chat being studied? If so, what characteristics of the neologism's progenitor predict the success of a new word within that initial social group? All in all, studying language development at such a scale can have implications for macro-level language change over time.

2. Data

The dataset is an iMessage group chat exported into CSV format using iMazing. It contains 25,572 messages across 9 users over a span of about 15 months. The messages are highly informal and include slang, repeated characters, emojis, and internet-specific language.

Several limitations in the dataset affect the analysis. First, not all users are present for the entire duration of the chat, which creates uneven participation (one person joined late and one person left midway through the pertinent timeframe). Second, image, GIF, and sticker data are not included, meaning some forms of communication are lost.

Despite these limitations, the dataset is well-suited for this study because it reflects organic, high-frequency digital communication between users.

3. Tools

All analysis was conducted in Python 3. The primary external libraries used were:

- **NLTK**, specifically its English words corpus, which was used to filter out standard English vocabulary and isolate candidate neologisms.
- **SciPy**, which was used to compute Pearson correlation coefficients and associated p -values.

No machine learning, language models, or APIs were used. The entire pipeline is rule-based and relies on custom scripts for parsing and analysis.

4. Methodology

- I. The CSV file is processed to extract structured message objects, including timestamp, sender, message text, and reactions.
- II. Words within the group chat that are not found in the NLTK English dictionary are identified as candidate neologisms. These include slang, abbreviations, and group-specific terms. The experimenter manually looked at each

of these words and removed any that were typos or were otherwise trivial.

- III. For each candidate term, the originator is identified along with all subsequent adopters. Adoption timing and usage frequency were tracked for each user as well.
- IV. To quantify linguistic behavior, I defined four custom metrics to capture different roles in the creation and spread of group-specific language. These are not drawn from existing literature; I essentially made them up to operationalize “innovation” and “adoption” in a measurable way. All metrics are computed per user after tracking each term’s originator, adopters, and usage over time.
 - **Creativity:** number of distinct neologisms a user originated that were adopted by at least one other user. Only terms selected in step II were considered.
 - **Influence:** average diffusion impact of a user’s words, computed per word using $(\# \text{ of adopters} \times \log(\text{total occurrences}))$, then averaged across all words they created for each person.
 - **Zeal:** proportion of post-adoption uses contributed by the user, calculated as $(\text{user’s uses after first adoption}) / (\text{total uses by all users})$.
 - **Fortitude:** resistance to adoption, computed as $(\# \text{ of eligible neologisms not adopted}) / (\text{total eligible neologisms})$, where eligibility means the user was active after the term’s introduction.
- V. Four conversational leadership metrics were computed directly from the corpus of messages.
 - **Reply-Worthiness:** replies received per message, i.e., $(\text{total replies received}) / (\text{messages sent})$.
 - **Citations:** references from others, computed as $(\text{mentions} + \text{replies directed at the user}) / (\text{messages sent})$, excluding self-mentions.
 - **Cold-Starts:** proportion of messages that initiate new threads, defined as messages sent after ≥ 2 hours of inactivity, divided by total messages sent.
 - **Reaction-Causing:** reactions received per message, i.e., $(\text{total reactions received}) / (\text{messages sent})$, with reactions parsed from the raw text field.

All metrics are min-max normalized across users for comparability.

- VI. **Correlation analysis.** Pearson correlations are computed between all linguistic and leadership metrics.

5. Results

The main output of the analysis is a correlation matrix between linguistic and leadership metrics (Table 3). Given the small sample size (9–10 users depending on the exact time window), only correlations with $|r| \geq 0.63$ are statistically significant at $p < 0.05$.

The most notable result is Citations vs. Linguistic Influence ($r = -0.689$, $p = 0.028$, statistically significant). This

indicates a moderate negative relationship between how influential a user is linguistically and how often they are directly referenced by others. However, it should be noted that this finding involves user B, who has a citations value of 1.0 (normalized) and linguistic influence of 0.0 – a single extreme data point that almost certainly anchors the correlation.

Other observed trends include: reaction-causing vs. zeal, a moderate negative relationship (not significant); citations vs. creativity, a moderate negative relationship (not significant); and cold-starts vs. influence, a moderate positive relationship (not significant).

Overall, most relationships are not statistically significant and should be interpreted as exploratory. Full per-user results are listed in the Appendix (Tables 1 and 2).

6. Interpretation and Conclusion

The results suggest that linguistic innovation and conversational leadership are not strongly aligned. In particular, the only statistically significant finding indicates that users whose language spreads more widely are actually mentioned less often in conversation.

This implies that linguistic influence operates in a more diffuse manner. Instead of drawing attention to the originator, successful language embeds itself in the group’s shared vocabulary for other reasons, or due to qualities of the word itself as opposed to the characteristics of the first utterer.

Additionally, creativity does not appear to correlate strongly with any leadership metric. Users who create new language are not necessarily the ones receiving the most replies, mentions, or reactions. This suggests that producing neologisms does not equate to controlling the conversation.

7. Limitations

There are several limitations to this study. The sample size is very small, which limits statistical power and makes results sensitive to outliers. The dataset also lacks complete information, particularly for media-based communication and for non-unigram features such as two-word phrases.

Additionally, the neologism extraction process is imperfect. It is possible that many mistyped or alternatively-stylized words may not have been captured when they should have been.

Perhaps most importantly, the analysis does not account for real-world language activity outside the chat in real life. Many instances of linguistic innovation and adoption may have first occurred outside the digital medium, and many real-life interactions may have influenced conversational dynamics. Lastly, in the case of this group of friends, there was an almost-identical group chat (consisting of the same members) that preceded the studied group chat for almost 2 years on another platform, whose contents could have been – but were not – included in this study. Any neologisms that originated there would appear as simultaneous adoptions or mis-attributed originations in this data.

A. Appendix

Table 1: User-level conversational leadership metrics (raw and min-max normalized values).

User	Raw				Normalized			
	Reply-worth.	Citations	Cold-starts	Reaction-c.	Reply-worth.	Citations	Cold-starts	Reaction-c.
A	0.120936	0.044863	0.050715	0.481144	0.498211	0.000000	0.802603	0.432628
B	0.082645	1.983471	0.000000	0.454545	0.340464	1.000000	0.000000	0.387312
C	0.100274	0.124559	0.041520	0.231101	0.413091	0.041110	0.657079	0.006631
D	0.120970	0.093840	0.040536	0.365784	0.498351	0.025264	0.641514	0.236089
E	0.096658	0.060097	0.028558	0.227209	0.398193	0.007858	0.451950	0.000000
F	0.102502	0.190970	0.051556	0.267846	0.422267	0.075367	0.815906	0.069234
G	0.000000	0.497392	0.026082	0.246218	0.000000	0.233430	0.412772	0.032386
H	0.242741	0.335656	0.058072	0.814170	1.000000	0.150001	0.919030	1.000000
I	0.151724	0.334483	0.044828	0.351724	0.625045	0.149396	0.709427	0.212136
J	0.171594	0.229565	0.063188	0.485797	0.706902	0.095275	1.000000	0.440555

Table 2: User-level linguistic innovation metrics (raw and min-max normalized values).

User	Raw				Normalized			
	Creativity	Influence	Zeal	Fortitude	Creativity	Influence	Zeal	Fortitude
A	3	15.177550	0.070056	0.512195	0.272727	1.000000	0.215504	0.333333
B	0	0.000000	0.003509	0.902439	0.000000	0.000000	0.000000	0.925926
C	11	14.295390	0.072600	0.780488	1.000000	0.941878	0.223739	0.740741
D	4	13.345110	0.062918	0.487805	0.363636	0.879267	0.192388	0.296296
E	7	14.733430	0.199785	0.292683	0.636364	0.970739	0.635609	0.000000
F	6	7.321628	0.312309	0.439024	0.545455	0.482399	1.000000	0.222222
G	3	7.496292	0.135972	0.292683	0.272727	0.493907	0.428962	0.000000
H	3	13.904860	0.028683	0.682927	0.272727	0.916147	0.081523	0.592593
I	0	0.000000	0.012987	0.951220	0.000000	0.000000	0.030694	1.000000
J	4	13.623240	0.036810	0.731707	0.363636	0.897592	0.107840	0.666667

Table 3: Pearson correlations (r) between linguistic innovation and conversational leadership metrics, with p -values in parentheses. Bold indicates $|r| \geq 0.63$ ($p < 0.05$).

	Creativity	Influence	Zeal	Fortitude
Reply-worthiness	−0.120 (0.741)	0.261 (0.467)	−0.383 (0.274)	0.446 (0.196)
Citations	−0.533 (0.112)	−0.689 (0.028)	−0.352 (0.319)	0.448 (0.194)
Cold-starts	0.229 (0.524)	0.525 (0.119)	0.071 (0.845)	−0.017 (0.962)
Reaction-causing	−0.438 (0.206)	0.141 (0.697)	−0.534 (0.112)	0.348 (0.324)